Detection of CMB B-mode Polarisation: Experimental Challenges and Computational Methods

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Abstract

Modern cosmology has achieved great success with the hot Big Bang model. However, the flatness and horizon problems remain unresolved within this framework, motivating the theory of cosmic inflation. Inflation proposes a brief period of exponential expansion driven by a scalar field, explaining the observed large-scale isotropy and flatness. Quantum fluctuations during this epoch generated both scalar and tensor perturbations, leaving imprints on the Cosmic Microwave Background (CMB). While scalar perturbations are confirmed by temperature anisotropies and E-mode polarisation, tensor perturbations, manifested as primordial B-modes, remain undetected. Detecting such B-modes remains a central goal in modern cosmology. Many inflationary models predict that the tensor perturbation is about a thousand times weaker than the scalar fluctuations ($r \sim 10^{-3}$, the tensor-to-scalar ratio), whereas current observational analyses place an upper limit of $r \approx 0.03$. Future missions such as the *Simons Observatory* and *LiteBIRD* are expected to approach sensitivities at or slightly above this level. Experimental searches face major obstacles from instrumental systematics, gravitational lensing, and astrophysical foregrounds. Modern instruments have largely mitigated systematic errors, and delensing techniques show a promising outlook, shifting the main challenge to foreground removal. Beyond traditional component-separation methods, neural-network approaches have emerged as powerful tools for non-linear foreground modelling. Specifically, recent developments in spherical Bayesian neural network showed potential application in scientific evaluation. Historical trends suggested that reaching $r \sim 10^{-3}$ by 2040 is achievable, but signal-to-noise ratio analysis showed that thermal dust must be suppressed by $\gtrsim 97\,\%$ for a meaningful detection in a *LiteBIRD*-like experiment. This highlights both the importance of broad-frequency satellite measurements and the need for improved modelling of polarised dust emission.

1 Introduction

Cosmology is the scientific study of the origin, structure, evolution and fate of the Universe as a whole. It seeks to understand the fundamental laws governing the cosmos and to describe its large-scale properties [1]. The modern era of cosmology began in the early 20th century with Einstein's general relativity, which provided a framework where the Universe itself could evolve dynamically. Based on Einstein's' equations, Friedmann and Lemaître independently demonstrated that the Universe can be expanding or contracting [2,3]. This theoretical prediction was later confirmed observationally when Edwin Hubble discovered that galaxies are receding from us, with recession velocity proportional to distance [4]. This observation laid the foundation of modern cosmology, revealing that the Universe is expanding and that both space and time are finite.

Another observational milestone was the accidental discovery of the Cosmic Microwave Background (CMB) by Penzias and Wilson (1965), who detected an unexplained, isotropic microwave excess. It was soon recognised as relic radiation from the Big Bang by Dicke, Peebles, Roll and Wilkinson (1965), revealing that the Universe is filled with a nearly perfect 2.7 K blackbody radiation field [5,6]. This observation, together with Hubble's cosmic expansion, provided evidence for the hot Big Bang model that if the Universe is expanding today, then tracing its history backward implies that it was once much hotter and denser, originating from an extremely compact initial state.

Subsequent observations have refined this picture. Measurements of galactic rotation curves and gravitational lensing indicate that most of the Universe's mass is non-luminous dark matter. Observations of distant Type Ia supernovae revealed that the cosmic expansion is accelerating, leading to the introduction of a repulsive component termed "dark energy" [7,8]. Together with dark matter, this forms the ΛCDM model, in which the Universe comprises roughly 68 % dark energy (Λ), 27 % cold dark matter (CDM), and about 5 % ordinary matter (baryons). This model successfully reproduces the thermal and structural evolution of the cosmos as illustrated in Figure 1. While the ΛCDM model is adopted as the prevailing model in this paper, it is worth noting that more recent studies have challenged the accelerating Universe and the current ΛCDM framework, showing mild preference for a dynamical form of dark energy such as the w_0w_aCDM model [9,10].

According to the thermal history, the CMB originates from the epoch of recombination. Before recombination, the Universe was filled with a so-called photon–baryon fluid, in which photons were tightly coupled to baryons and unable to travel freely, making the Universe opaque. As the Universe expanded and cooled, free electrons and protons combined to form neutral hydrogen, allowing photons to decouple from matter. These photons became the last scattered light, meaning CMB is the oldest observable light in the Universe. However, the current observations

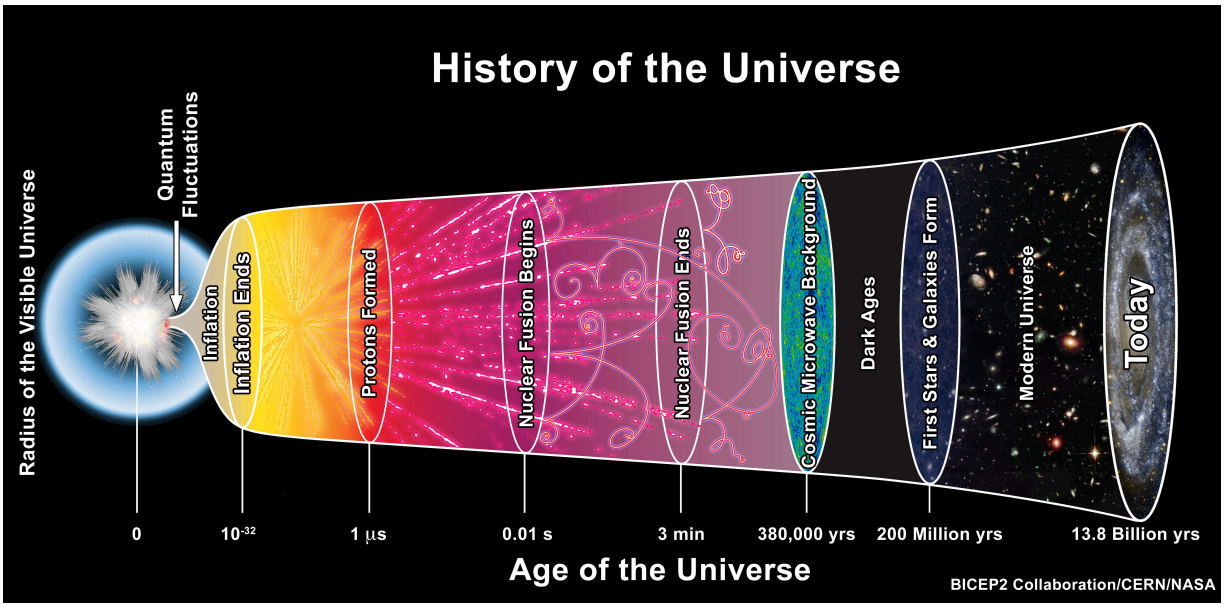

Figure 1: Timeline of the Universe. Illustration showing the evolution of the Universe from the Big Bang to the present day. The horizontal axis represents cosmic time on a logarithmic scale. The vertical direction denotes the approximate physical scale. The Cosmic Microwave Background (CMB) originates from the epoch of recombination, approximately 380,000 years after the Big Bang [11].

still leave several problems unresolved, most notably the flatness problem and horizon problem. To address these issues, cosmologist proposed the theory of cosmic inflation, an early epoch of exponential expansion in the very early Universe, driven by a hypothetical scalar field known as the inflaton.

Inflation predicts that quantum fluctuations of the inflaton field generated both scalar and tensor perturbations, leaving imprints on the CMB. The scalar perturbations, responsible for temperature anisotropies and E-mode polarisation, are now strongly supported by precise CMB observations. In contrast, the tensor perturbations remain undetected. An observational evidence of these tensor perturbation would be B-mode polarisation pattern in the CMB, produced by primordial gravitational waves under the assumption of quantised gravity. Therefore, detecting such B-modes remains a central goal in modern cosmology. It serves not only as a critical test of the inflationary paradigm, but also as a potential first observational evidence of quantised gravity, offering crucial insight into reconciling the two foundational pillars of modern physics — general relativity and quantum mechanics.

In this dissertation, we explore the motivation for inflation and examine current experimental efforts. We also discuss how modern computational techniques, particularly neural network based, can contribute to the future detection of primordial B-mode.

2 Theory

2.1 Cosmological Framework

The cosmological principle states that the Universe is homogeneous and isotropic on large scales. A general curved spacetime with these symmetries is described by the Friedmann-Robertson-Walker (FRW) metric,

$$ds^2 = -c^2dt^2 + a^2(t)\left(\frac{dr^2}{1-kr^2} + r^2d\Omega^2\right) \tag{1}$$

where $d^2\Omega$ is the solid angle element, c is the speed of light, r is the comoving distance, a is the scale factor. The spatial curvature k take values 0, ± 1 corresponding to flat(0), closed(+1) and open(-1) geometries respectively. The physical distance is given by $D(t) = a(t)r$, while the comoving distance r represents invariant distance between objects under the expansion of the Universe. Assuming the space is perfect fluid, substituting this metric into Einstein's field equations yields the Friedmann equation,

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G\rho}{3} - \frac{kc^2}{a^2}\,. \tag{2}$$

By differentiating the Friedmann equation with respect to time, one can obtain the acceleration equation,

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3\rho}{c^2}\right)\,. \tag{3}$$

Another key relation is the fluid equation,

$$\dot{\rho} + 3\frac{\dot{a}}{a}\left(\rho + \frac{\rho}{c^2}\right) = 0 \tag{4}$$

see Sec. 1.5 of Ref. [12].

These three equations form the fundamental framework of cosmology, describing the large-scale dynamics of the Universe under the assumptions of homogeneity and isotropy.

2.2 Flatness Problem

To describe flatness problem, it is useful to define critical density

$$\rho_c = \frac{3H^2}{8\pi G} \tag{5}$$

which represent spatially flat Universe, $k = 0$. With this definition, Friedmann equation can be rewritten as

$$(\Omega^{-1} - 1)\rho a^2 = \frac{-3kc^2}{8\pi G} \tag{6}$$

where $\Omega = \rho/\rho_c$ and the right hand side is constant [13].

During the radiation and matter dominated expansion following recombination, the energy density scales as $\rho \propto a^{-4}$ (radiation) and $\rho \propto a^{-3}$ (matter). Hence, as the Universe expands, ρa^2 decreases and consequently $(\Omega^{-1} - 1)$ increases.

However, current observations show that Ω is extremely close to unity as $\Omega = 1.0004 \pm 0.0018$, according to *Planck* data [14]. This implies that, when we trace the time backward, Ω must have been even closer to one, as though the Universe began with the the critical density to extraordinarily high precision. This is known as the fine-tuning problem or flatness problem, suggesting that early Universe appeared to start with a very specific density without any obvious reason.

2.3 Horizon Problem

The horizon problem concerns the lack of causal connection across the observable Universe. The CMB is remarkably uniform, with temperature fluctuation of only $\Delta T \approx 10\,\mu K$, establishing perfect blackbody spectrum [15, 16]. This high degree of isotropy implies that distant regions of the early Universe were in thermal equilibrium. However, such regions could not have been in causal contact at the time of recombination, since the size of the causal horizon then corresponded to only about one degree on today's CMB sky. This can be qualitatively understood by considering the radius of causal horizon, which defines the maximum region where interaction can occur, limited by the speed of light

$$r_H = \frac{R_H}{a} = \frac{1}{Ha} \tag{7}$$

where r_H denotes the Hubble radius, describing comoving radius of the causal horizon in natural units. Its time derivative is

$$\dot{r}_H = -\frac{1}{a}\left(\frac{\dot{H}}{H^2} + 1\right) . \tag{8}$$

For matter dominated expansion, the scale factor follows $a \propto t^{2/3}, \dot{a} \propto \frac{2}{3}t^{-2/3}$, which gives $\dot{r}_H > 0$. Thus, as we trace the evolution backward in time, r_H decreases, meaning the causal horizon in the early Universe was much smaller than the angular size of the current CMB sky.

2.4 Inflationary Dynamics

Inflation provides a mechanism to resolve the major issues of the standard Big Bang cosmology, particularly the flatness and horizon problems. It postulates the existence of a hypothetical scalar field known as the inflation, which dominated the energy density of the early Universe. If this field evolved under a sufficiently flat potential, a condition known as the "slow-roll" condition, it would have driven a brief period of exponential expansion in the very early Universe. This rapid expansion naturally explains both the flatness of space and the large-scale causal connectivity observed in CMB.

In addition, when the inflaton field eventually rolls down to the minimum of its potential, it begins to oscillate about the minimum and decays into ordinary matter and radiation, known as reheating process.

This can be understood qualitatively by considering the energy density associated with the inflaton field expressed as

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi) \tag{9}$$

where ϕ is the inflaton field and $V(\phi)$ is the potential energy. Under the slow-roll condition, the kinetic term is negligible compared to the potential energy, i.e. $\dot{\phi}^2 \ll V(\phi)$, resulting in an approximately constant energy density, $\rho_\phi \approx V(\phi)$. Substituting this into Equation 2 yields

$$H^2 \approx \frac{8\pi G}{3} V(\phi) \tag{10}$$

where the Hubble parameter H is nearly constant during inflation. Consequently, the scale factor grows exponentially as

$$a(t) = a_0 e^{H(t-t_0)} \tag{11}$$

where $H = \dot{a}/a$ by definition.

During this phase, since ρ remains approximately constant while a^2 increases exponentially, the term $(\Omega^{-1} - 1)$ in Equation 6 is driven toward zero, forcing Ω to be one. In this way, inflaton field effectively resets the curvature as $\Omega = 1$ by diluting any initial deviation from flatness.

Moreover, inflation naturally resolves the horizon problem as well. During the exponential expansion, the Hubble radius evolves as $\dot{r}_H = -\frac{1}{a} \propto -e^{-Ht}$. Tracing the evolution backward in time, r_H increases exponentially, meaning that regions of the Universe that now appear causally disconnected were once within the same causal patch before inflation. In other words, inflation stretched these initially interacting regions beyond the current observable horizon, explaining the uniformity of the CMB temperature across the sky.

2.5 Primordial Perturbations from Inflation

The inflation is assumed to be quantised and therefore follows the framework of quantum field theory. Its fluctuation can be described as a linear perturbation,

$$\phi(t, \vec{x}) = \bar{\phi}(t) + \delta\phi(t, \vec{x}) \tag{12}$$

where $\bar{\phi}(t)$ denotes the spatially averaged background field and $\delta\phi(t, \vec{x})$ represents its perturbation, see Sec. 6.5.1 of Ref. [17]. The evolution of each Fourier mode of the fluctuation is governed by

$$\delta\ddot{\phi}_k + 3H\delta\dot{\phi}_k + \frac{k^2}{a^2}\delta\phi_k = 0 \tag{13}$$

where k is the comoving wavenumber. During inflation, $\frac{k}{a} \ll H$, therefore Equation 13 simplifies to

$$\delta\ddot{\phi}_k + 3H\delta\dot{\phi}_k \approx 0 \tag{14}$$

which describes a heavily damped oscillator. Consequently, the fluctuations become effectively "frozen", see Sec. 2.2 of Ref. [18]. After inflation ends, these frozen inflation fluctuations become primordial perturbation in energy density during reheating, serving as the seeds of large-scale structure formation.

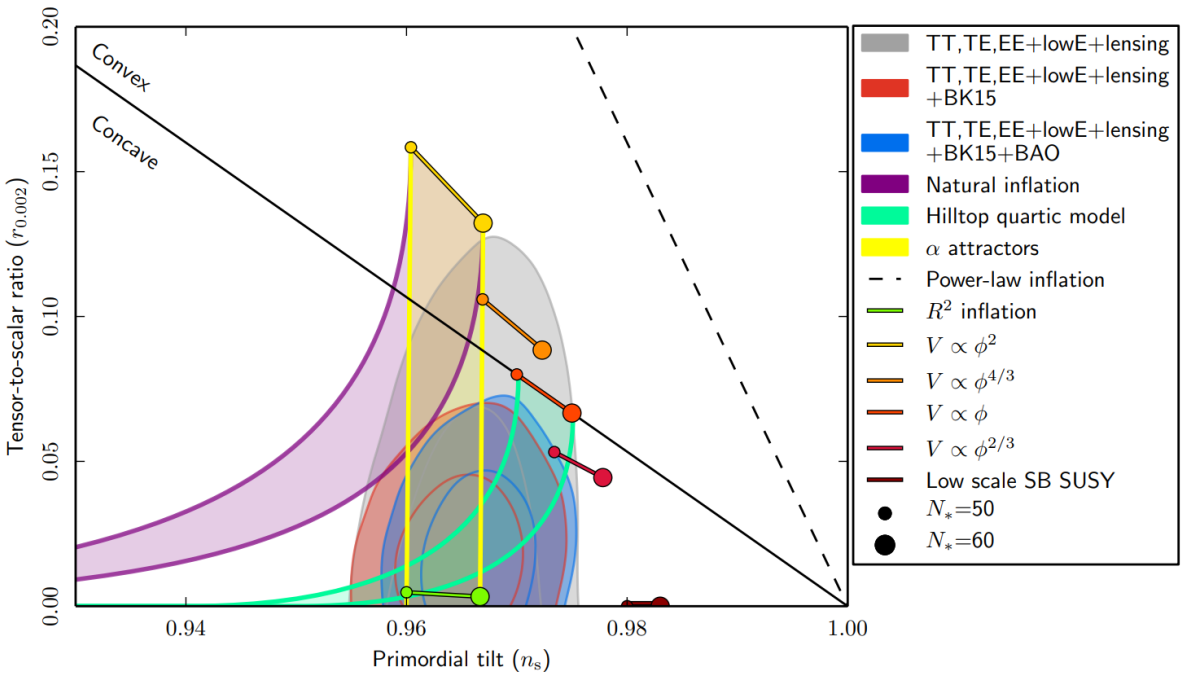

Figure 2: Theoretical predictions of the scalar spectral index n_s and tensor-to-scalar ratio r for selected inflationary models. The shaded contours show the marginalized joint 68 % and 95 % CL regions at the pivot scale $k = 0.002\,\mathrm{Mpc}^{-1}$, derived from combinations of Planck temperature and polarisation data with BICEP/Keck (BK15) and BAO measurements. These measurements already begin to rule out portions of the parameter space associated with several inflationary models. Reproduced from Planck Collaboration (2020) [19]

Inflation also predicts the existence of tensor perturbations, arising from quantum fluctuations of the spacetime metric itself under the assumption of quantised gravity. These perturbations can be expressed as

$$ds^2 = a^2(\eta)[-d\eta^2 + (\delta_{ij} + h_{ij})dx^i dx^j] \tag{15}$$

where η is conformal time defined by $d\eta = dt/a(t)$ and h_{ij} denotes the perturbation in space metric, see Sec. 18 of Ref. [20]. Analogous to the scalar case, these tensor modes are stretched beyond the horizon during inflation and become frozen. As it re-enters the horizon at later epochs, they become gravitational waves propagating through spacetime, producing oscillatory motions in the photon-baryon fluids.

Since the Universe is statistically homogeneous and isotropic, the CMB temperature and polari-

sation fields have no preferred position or direction on the sky. Thus, the homogeneity (translational symmetry) allows CMB perturbations to be decomposed into Fourier modes, while isotropy (rotational symmetry) ensures that the statistics only depend on the magnitude of wavevector $k = |\vec{k}|$, see Sec. 5.2 of Ref. [12].

Accordingly, the perturbation can be characterised by their power spectra $P(k)$, which describes the variance of mode amplitudes as a function of wavenumber. The scalar and tensor power spectra are commonly parameterised as

$$P_s(k) = A_s \left(\frac{k}{k_*}\right)^{n_s - 1}, \qquad P_t(k) = A_t \left(\frac{k}{k_*}\right)^{n_t} \tag{16}$$

where A_s and A_t are the respective amplitudes evaluated at the pivot scale k_* (typically chosen as $0.05\,\mathrm{Mpc}^{-1}$ or $0.02\,\mathrm{Mpc}^{-1}$), and n_s and n_t are scalar and tensor spectral indices, see Sec. 9.3 of Ref. [21]. Most inflationary models predict a nearly scale-invariant scalar spectrum, $n_s \lesssim 1$, as illustrated in Figure 2.

The amplitude of the tensor perturbation is conventionally expressed in terms of the tensor to scalar ratio

$$r \equiv \frac{P_t(k_*)}{P_s(k_*)}\,. \tag{17}$$

Different slow-roll models predict distinct relationships between n_s and r, defining characteristic trajectories in the (n_s, r) plane. Typical inflationary models predict values of r ranging from 0.001 to 0.1, as shown in Figure 2, where the shaded regions denote the 95% confidence limit from current observation.

2.6 CMB Anisotropy and Polarisation Power Spectra

Primordial perturbations generated during inflation lead to observable quantities, temperature anisotropy, E-mode polarisation and B-mode polarisation in CMB. The E-mode polarisation forms a gradient-like pattern of linear polarisation, while the B-mode polarisation forms curl-like pattern as illustrated in Figure 3.

The temperature anisotropies are direct result of scalar perturbations that produce fluctuation in the photon temperature through gravitational red shifts (SachsWolfe effect) [23]. Primordial E-mode polarisation arises from the scalar perturbation as well. The velocity field associated with scalar density fluctuation induces a local quadrupole anisotropy in the radiation field, which creates E-mode linear polarisation through Thomson scattering, see Secs. 10.5–10.8 of Ref. [17]. In contrasts, primordial B-mode polarisation arises from primordial gravitational waves, which is tensor perturbation. At recombination, these time varying anisotropies induced both E-mode and B-mode both via Thomson scattering, see Sec. 10.9 of Ref. [17].

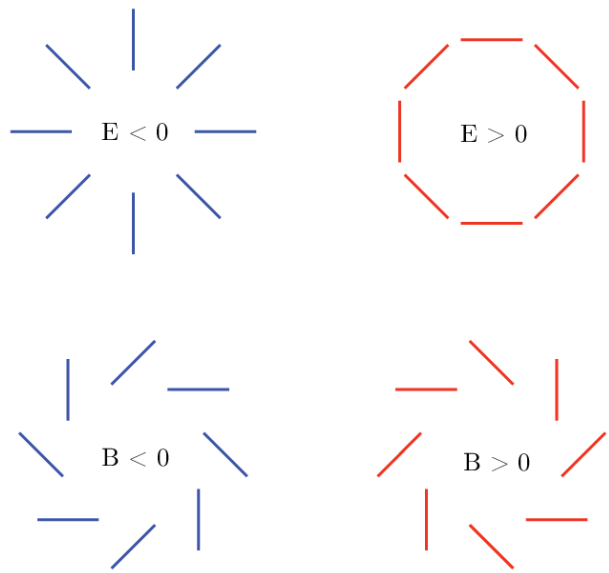

Figure 3: Illustration of the polarisation patterns associated with E- and B-modes. E-modes (top) exhibit gradient-like patterns, while B-modes (bottom) display curl-like patterns. Reproduced from Baumann et al. (2009) [22].

These physical observables can be quantified as

$$\begin{aligned} C_\ell^{TT} &= 4\pi \int \frac{dk}{k}\, P_s(k)\, |\Delta_\ell^T(k)|^2, \qquad & C_\ell^{EE} &= 4\pi \int \frac{dk}{k}\, P_s(k)\, |\Delta_\ell^E(k)|^2, \\ C_\ell^{BB} &= 4\pi \int \frac{dk}{k}\, P_t(k)\, |\Delta_\ell^B(k)|^2, \qquad & C_\ell^{TE} &= 4\pi \int \frac{dk}{k}\, P_s(k)\, \Delta_\ell^T(k) \Delta_\ell^E(k) \end{aligned} \tag{18}$$

where C_ℓ^{TT}, C_ℓ^{EE} and C_ℓ^{BB} represent an angular power spectra of the CMB temperature anisotropy, E-mode and B-mode polarisation, describing the strength of fluctuations on angular scale corresponding to the multipole index ℓ. The cross power spectrum C_ℓ^{TE} quantifies the correlation between temperature anisotropies and E-mode polarisation. The term $\Delta_\ell^{T,E,B}(k)$ denotes the corresponding transfer functions, which describe how each primordial perturbation mode of wavenumber k evolves into each observables and how it is projected onto the observed sky at multiple ℓ, see Sec. 8.5.2 of Ref. [17]. Together, these spectra capture the statistical properties of CMB anisotropies predicted by inflation, depicted in Figure 4.

The TT spectrum shows the initial condition set by inflation at larger angular scales(low ℓ), including the Sachs-Wolfe plateau arising from gravitational potential fluctuations [23]. At smaller angular scales (high ℓ), it shows a series of acoustic peaks, which is a direct result of primordial scalar perturbation, see Sec. 8 of Ref. [17]. The detailed shape of these peaks arise from physical process beyond inflation such as Silk damping tail [24]. The first detection of this pattern was detected by *BOOMERanG*, *MAXIMA*, and further confirmed by *WMAP* [25–27]. The *Planck* refined these results and confirmed the nearly scale-invariant scalar spectrum with

a spectral index of $n_s = 0.9649 \pm 0.0042$ [28]. These results strongly supports the inflationary model. The TE cross-correlation exhibits alternating positive and negative correlation. This is

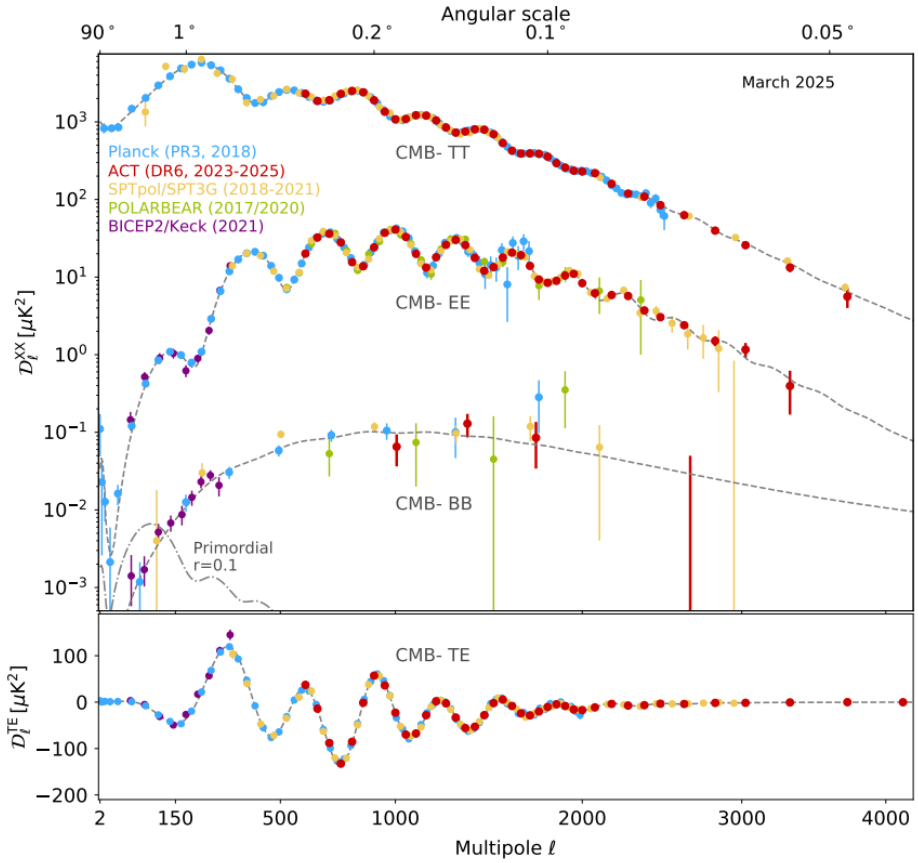

Figure 4: The figure summarises the current observational status of CMB temperature and polarisation anisotropies across a wide range of angular scales. The dashed line represents the best fit ΛCDM model to P-ACT data, compared with other CMB experiments. For comparison, a primordial gravitational-wave B-mode template with tensor-to-scalar ratio $r = 0.1$ is shown (grey dot-dashed line), illustrating the expected peak at $\ell \approx 80$. Reproduced from ACT Collaboration [29].

because temperature fluctuations trace the density of photon-baryon plasma perturbation while the polarisation is sourced by the velocity field of the same plasma, the two are approximately $\pi/2$ out of phase. Similarly, the auto-correlation of E-mode polarisation, EE spectrum, shows alternating peaks and troughs with half period out of phase with the TT spectrum. At the very low multipoles ($\ell \lesssim 10$), both TE and EE spectra contain reionisation bump, produced when reionised electron at later epochs scatter the CMB photons again and regenerate large-scale polarisation [24, 30]. These feature also have been measured with precise agreement with theory. *WMAP*, firstly detected the expected sign changing patter in the TE correlation. *Planck* confirmed this with higher precision showing the predicted phase shift and amplitude ratio between TT and EE [31, 32]. More recent ground-based experiments such as *Atacama Cosmology Telescope (ACT)* and *SPTpol* have provided high resolution measurements at small angular scales, further supporting the inflationary model [33, 34].

The BB power spectrum shows auto-correlation of B-mode polarisation. The primordinal B-mode spectrum predicted by inflation shows two distinct features. The first is a broad bump at the very low multipoles ($\ell \lesssim 10$) produced during reionisation, the same as the TE, TE spectra. The second bump is at $\ell \approx 80$, corresponding to the recombination bump produced by tensor perturbation at the surface of the last scattering. With the corresponding tensor to scalar ratio r from Figure 2, most of theories expect B-mode power sepctra C_ℓ^{BB} signal of order 0.1–0.01 $\mu\mathrm{K}^2$ at multipole $l \approx 80$, see Sec. 5 of Ref. [18]. This B-mode power spectrum is not yet measured precisely due to many limitation such as lensing, instrumental noise and foreground objects which produce B-mode polarisations as well.

3 Experimental and Computational Approaches to B-mode Detection

3.1 Polarisation Measurement and Harmonic Analysis

The linear polarisation of the CMB is characterised by the Stokes parameters Q and U which describe the intensity difference between orthogonal polarisation directions. For a given sky direction $\hat{n}$ and polariser orientation angle ψ, the measured intensity is expressed as

$$d(\psi, \hat{n}) = I(\hat{n}) + Q(\hat{n})\cos(2\psi) + U(\hat{n})\sin(2\psi) \tag{19}$$

where d is the detector signal and $I(\hat{n})$ is the total intensity. By rotating the polarisation angle ψ during measurement, one can extract the values of Q and U for each direction on the sky.

Since Q and U depend on the choice of coordinate basis, it is convenient to combine them into a complex spin-2 field as

$$P_\pm(\hat{n}) = Q(\hat{n}) \pm i, U(\hat{n}) \tag{20}$$

and can be expanded in terms of spin-weighted spherical harmonics ${}_{\pm 2}Y_{lm}$

$$P_\pm(\hat{n}) = \sum_{\ell m} a_{\pm 2,,\ell m}, {}_{\pm 2}Y\ell m(\hat{n}) \tag{21}$$

where $a_{\pm 2,lm}$ are the corresponding harmonic coefficients. From these coefficients, one can define two scalar quantities representing E-mode and B-mode components of the polarisation field

$$a_{\ell m}^E = -\frac{1}{2}\left(a_{+2,,\ell m} + a_{-2,,\ell m}\right), \qquad a_{\ell m}^B = \frac{i}{2}\left(a_{+2,,\ell m} - a_{-2,,\ell m}\right) \tag{22}$$

where $a_{\ell m}^{E,B}$ represents the spherical harmonic coefficient of each mode. Finally, one can find power spectra of each mode

$$C_\ell^{EE} = \frac{1}{2\ell+1}\sum_m \left|a_{\ell m}^E\right|^2, \qquad C_\ell^{BB} = \frac{1}{2\ell+1}\sum_m \left|a_{\ell m}^B\right|^2. \tag{23}$$

These relations follow the standard formalism of CMB harmonic analysis, see Secs. 4–5 of Ref. [18].

Modern CMB polarimeters measure the linear polarisation of the microwave sky by modulating and detecting the Stokes parameters Q and U across multiple frequency bands. These can be broadly categorised into three types: ground-based, balloon-borne, and satellite platforms. While each serves a different purpose, they share a common architecture, including telescope optics, polarisation modulators and cryogenic bolometric detector arrays.

The telescope optics collect and focus incoming radiation onto the detectors. They may employ refracting optics for small aperture (e.g, BICEP/*Keck*), which provide simple beam, or reflecting optics for larger aperture systems (e.g, *POLARBEAR*, *Simons Observatory*) [35–37]. Polarisation modulation is the core mechanism for measuring Q and U. This is typically achieved either through a rotating half-wave plat (HWP) or by pair differencing. THE HWP consists of birefringent crystal which continuously rotates the incoming polarisation (used in *POLARBEAR-2*, *LiteBIRD*, *SPIDER*), while pair differencing measures the signal difference between two orthogonally oriented detectors (*BICEP/Keck*) [38–41]. Finally, the detectors absorb the modulated radiation and convert it into electrical signals. In most modern instruments, transition-edge sensors (TES) or Kinetic Inductance Detectors (KID) are commonly used.

In CMB polarisation experiments, several instrumental systematics dominate the risk of contamination of the B-mode signal. First, telescope optics can introduce intensity to polarisation leakage, where beam mismatch or differential gain leads to false E and B-mode signals. In addition, instrumental polarisation arising from non-ideal reflections or refraction can produce secondary polarised signals from unpolarised sources [42]. Secondly non-ideal polarisation modulators, particularly for half-wave plates (HWP) can introduce frequency dependent phase shift [43]. Finally at the detector level, thermal fluctuation in superconducting sensors (TES or KID) can change the gains, resulting in false change in output current or voltage [44]. Because the primordial B-mode signal is extremely faint, even small systematic uncertainties can have large impact on its detection.

Modern CMB experiments mitigate these effects through a combination of instrument design, calibration and data analysis. For example, in *BICEP2*, instrumental leakage was mitigated through "deprojection" techniques that removed temperature-to-polarisation coupling from beam mismatch [45]. For experiments employing HWPs, it was demonstrated that the effect of ambient temperature on a continuously rotating HWP can be effectively mitigated, resulting in a specific impact on the tensor-to-scalar ratio of $\delta r < 0.001$ [46].

Overall, systematic errors in modern instruments are now considered to be well controlled and are no longer the dominant obstacle in detecting primordial B-modes, compared with astrophysical

foregrounds and gravitational lensing contamination.

3.2 Lensing Contamination and Delensing Techniques

3.2.1 Lensing Mechanism and Contamination

Lensing is an effect of gravitational deflection of photons by large-scale-structures(LSS). Figure 5 illustrates how the gravitational potentials from such LSS can deflect CMB photons and remap the CMB sky. It causes part of the E-mode pattern to be converted into B-modes. The B-mode signal from gravitational lensing dominates over the primordial component at multipoles $\ell > 1000$ acting as a foreground floor with an amplitude of the order of $0.01\,\mu\mathrm{K}^2$ [47]. Thus to measure primordial B-mode, lensing must be accounted.

Lensing of observed Stoke's parameter Q and U can be descried as

$$Q(\hat{n}_{\mathrm{obs}}) = Q(\hat{n} + \vec{d}), \qquad U(\hat{n}_{\mathrm{obs}}) = U(\hat{n} + \vec{d}) \tag{24}$$

where $\hat{n}_{\mathrm{obs}}$ describes the observed coordinates on the sphere, $\hat{n}$ descries the original coordinates, and $\vec{d}$ describes the displacement on the sphere.

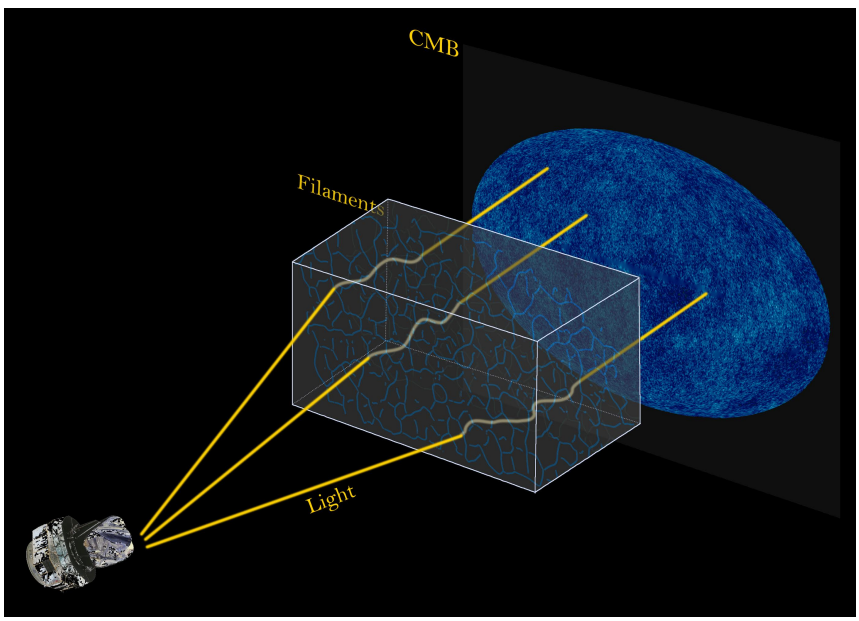

Figure 5: Illustration of how large-scale cosmic filaments deflect the paths of photons from the cosmic microwave background. Light from the CMB (right) is gravitationally distorted as it passes through the filamentary structure of the cosmic web (centre) before reaching the Planck satellite (left). Credit: Siyu He, Shadab Alam, Wei Chen, and Planck/ESA; adapted from [48].

Since $\vec{d}$ is 2D vector field, it can be described witt general potential

$$\vec{d} = \nabla\phi + \nabla \times \vec{\Omega} \simeq \nabla\phi \tag{25}$$

where ϕ is general scalar potential and Ω is general vector potential. Since the curl-like component is negligible, $\vec{d}$ can be approximated with the scalar potential. Thus in principle, by knowing the potential ϕ, the mapped observed CMB can be undone, or delensed [49]. Here, ϕ can be interpreted as an integrated gravitational potential along the ling of sight, projected on the sphere, which can be formulated as

$$\phi(\hat{n}) = -2 \int_0^{\chi^*} d\chi \, W^{\phi}(\chi) \, \Phi(\chi\hat{n}, \eta_0 - \chi) \tag{26}$$

where χ is comoving distance, χ^* is the comoving distance CMB photon travels to reach us from recombinaton, $\Phi(\chi\hat{n}, \eta_0 - \chi)$ is the gravitational potential at the specific spacetime and $W^{\phi}(\chi)$ is the weight function describing how efficiently matter at distance χ lenses the CMB photon [50].

3.2.2 Delensing and Reconstruction Techniques

The most profound way of delensing is CMB-only reconstruction. Any CMB temperature or polarisation field on the sky can be expanded in spherical harmonics,

$$X(\hat{n}) = \sum_{l,m} X_{lm} Y_{lm}(\hat{n}) \tag{27}$$

where X is arbitrary field and Y_{lm} denotes the spherical harmonic functions. For the unlensed CMB, the multipoles are Gaussian and statistically isotropic. This implies that the covariance of the harmonic components of two fields X and X' is

$$\langle X_{\ell m}, X_{\ell' m'} \rangle = \delta_{\ell\ell'} \delta_{mm'} C_{\ell}^{XX'} \tag{28}$$

where $C_{\ell}^{XX'}$ is the the cross-power spectrum between the fields X and X'(T, E, B). However, gravitational lensing breaks this isotropy and mixes different multipoles. These correlations between different multipoles are directly related to lensing potential. Thus by measuring them, one can reconstruct a lensing potential, see Okamoto & Hu (2003) for full details [51]. Using this method, 7 % reduction in the B-mode polarisation power was achieved by Carron et al. in 2017 and recent work achieved 40 % reduction in the uncertainty of tensor-to-scalar ratio r with CMB-only reconstruction [52, 53].

However, in practice the accuracy of CMB-only reconstruction is limited by detector noise and foreground contamination [54]. A complementary approach is to use external tracers of large-scale structure to reconstruct the lensing potential. In principle, objects observed at different cosmic epochs can be used to probe distinct segments of the line-of-sight integral, thereby enabling a more accurate reconstruction of the lensing potential $\phi(\hat{n})$. At late times (low redshift), gravitational lensing of the CMB arises from large-scale structure traced by external observables such as galaxy clusters or the cosmic infrared background (CIB), which are statistically correlated with the CMB lensing potential [49, 55]. Cross-correlating these tracers with the CMB therefore provides an additional estimate of lensing potential, which can be combined with

CMB-only reconstruction to improve the delensing performance. For example, Yu et al. (2017) achieved a delensing factor of approximately 45 %, and recent work reports a 60 % reduction in the uncertainty of the tensor-to-scalar ratio r [53].

Future missions are expected to deliver substantial improvements in delensing performance. The *CMB-S4* experiment forecasts an effective removal of approximately 92–93 % of the lensing B-mode variance [56]. Similarly, the *LiteBIRD* mission anticipates an improvement of about 20 % in the constraint on r when its data are combined with that of *CMB-S4* [57]. These forecasts suggest a promising outlook for future delensing capabilities and the detection of primordial gravitational waves.

3.3 Foreground Emission and Component Separation Methods

3.3.1 Foreground Mechanisms

However, foreground emissions remain the major obstacle, as they exhibit highly complex and spatially varying patterns across the sky, closely linked to the Galactic magnetic field. Unlike lensing, which is theoretically well understood, foreground emissions lack a solid theoretical foundation and rely on empirical templates, making it extremely difficult to distinguish them from the primordial B-mode signal.

One of the Galactic foregrounds contaminating CMB observation is synchrotron radiation, arising from relativistic cosmic-ray electrons spiralling around the Galactic magnetic field lines. These cosmic-ray electrons are thought to originate from supernova remnants, pulsars and can be either compact or diffusive on the sky. As it moves perpendicular to the galactic magnetic field, it oscillates on the surface perpendicular to the magnetic field, and it naturally creates linearly polarised light. Due to the irregularities in the galactic magnetic field lines, the polarisation angles vary across the sky, leading to non-zero curl in the polarisation pattern, which is B-mode [58].

The spectral energy distribution of these electrons are dominated in the lower frequency (MHz) and in the frequency higher frequency range (the range of frequency for CMB) follows power law with power equals to $\beta \approx -3$ [59, 60]. Also its polarisation power spectrum follows power law $C_\ell \propto \ell^\alpha$, $\alpha \approx -2.5$ [59]. This information enables parametric modelling of synchrotron emission and is explicitly used in the BICEP/*Keck* foreground component separation likelihood [61]. In addition, its spatial morphology is correlated in large scale structure such as filamentary structures and the polarisation angle varies smoothly [62,63]. These characteristics make synchrotron emission distinguishable from the primordial B-mode signal through multi-frequency component separation. The BICEP/*Keck* Collaboration showed that the fitted synchrotron amplitude is tightly constrained and not detected in any multipole range, demonstrating that the multi-frequency pipeline removes synchrotron exceptionally well [64]. It is also found that the ratio

$C_\ell^{BB}/C_\ell^{EE} = 0.5$, indicating that the synchrotron polarisation field contains more E-mode than B-mode structure [65].

Another major foreground emission is thermal dust emission. The interstellar dust grains are thought to be composed of a mixture of amorphous silicate, graphitic grains, and polycyclic aromatic hydrocarbons (PAHs), with sizes ranging from sub-micron to angstrom scales, although their precise composition remains uncertain [66]. These grains absorb ultraviolet and optical radiation from nearby stars and re-emit thermally in the infrared and far-infrared at an equilibrium temperature of approximately 19.6 K, well described by a modified blackbody spectrum [67, 68].

Figure 6 illustrates how elongated dust grains align with the Galactic magnetic field and produce thermal emission polarised perpendicular to the field. The grains' irregular shapes allow them to acquire angular momentum from anisotropic radiation fields. This leads to aligning their shorter rotational axes with the local magnetic field. This process, known as radiative torque alignment (RTA), produces a net alignment of grains within the interstellar medium [69].

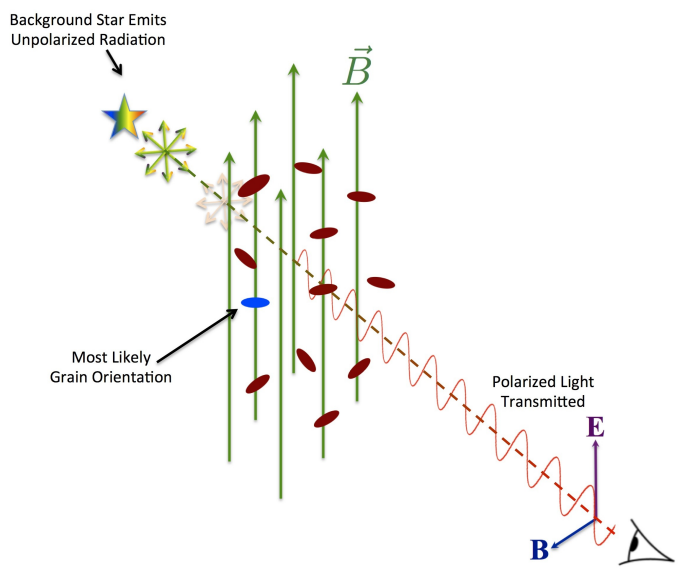

Figure 6: Illustration of the mechanism behind polarised thermal dust emission in the interstellar medium. Elongated dust grains align their short axes with the local Galactic magnetic field through radiative–torque processes, causing their thermal emission to be linearly polarised with a direction orthogonal to the projected magnetic field, whereas transmitted starlight becomes polarised parallel to it. Reproduced from Ref. [70].

Also due to their irregular shape, they posses anisotropic dielectric properties, and the aligned grains emit linearly polarised thermal radiation, with the polarisation direction orthogonal to the magnetic field [71]. Spatial variations in the Galactic magnetic field then imprint a pattern of E- and B-mode polarisation on the sky. Planck measurements show that the dust polarisation power spectra follow a power law $C_\ell \propto \ell^\alpha$ with $\alpha = -2.42 \pm 0.02$ [72]. As with synchrotron emission, this parametrisation is useful for constructing foreground models for component sepa-

ration and is explicitly employed in the BICEP/*Keck* likelihood [61]. In addition, the observed ratio $C_\ell^{BB}/C_\ell^{EE} = 0.5$ reflects the spatial morphology of polarised thermal dust emission, which is dominated by E-mode structure [72].

The B-mode amplitude from galactic dust emission reaches approximately $0.01\,\mu\mathrm{K}^2$ over $40 < \ell < 120$, coinciding with the multipole range of the primordial recombination bump [72]. Moreover, its modified blackbody emission overlaps with that of the primordial B-modes above 70 GHz. Consequently, polarised galactic dust emission constitutes the most significant astrophysical foreground emission, limiting the detection of the primordial B-mode signal. Therefore, accurate separation requires prior knowledge of physical properties of interstellar dust grains and careful multi-frequency analysis [73].

A well-known example of difficulty and the necessity of multi-frequency analysis is the *BICEP2* misinterpretation in 2014 [74]. The *BICEP2* collaboration initially reported the detection of a primordial B-mode signal with $r = 0.20^{+0.07}_{-0.05}$. However, subsequent analysis revealed that this result was not of cosmological origin. The experiment operated primarily at single frequency band (150 GHz) and significantly underestimated the contribution of Galactic dust emission on B-mode signals. Later studies that the observed signal could be entirely explained by Galactic dust emission, underscoring the importance of multi-frequency analysis [75].

3.3.2 Removal Techniques

There are several approaches for modelling and separating foreground components in CMB data, broadly classified as blind, semi-blind and non-blind (parametric) in terms of the degree of physical modelling they assume. Blind methods minimise physical assumptions and extract components statistically, whereas non-blind methods rely on explicit physical models. In terms of model fitting, the main strategies include linear combination methods, likelihood based methods, maximum entropy methods etc [76]. Each class has its own advantages and limitations, but two representative examples are discussed here, *Commander* and SMICA.

Commander, whose latest version is *Commander3*, is a Bayesian analysis framework that employs Gibbs sampling to jointly estimate astrophysical and instrumental parameters in the pixel domain [77,78]. *Commander* assumes a physical model for each foreground component and uses Bayesian sampling to estimate the full posterior probability distribution of all unknown parameters. From this joint posterior, any subset of parameters, such as component amplitudes, spectral indices or CMB power spectrum, can be estimated. This Bayesian framework offers great physical interpretability and rigorous uncertainty quantification, as it provides the complete posterior distribution of the model parameters. However, it is computationally demanding and inherently model-dependent, requiring well motivated physical priors [76]. *Commander* has been extensively used in the *Planck* mission analysis [79, 80].

Another major algorithm SMICA(Spectral Matching Independent Component Analysis) is a semi-blind likelihood-based covariance-matching method which operates in the power spectrum domain [81]. Unlike *Commander*, it does not use explicit physical models of the foregrounds. Instead it estimates model parameters by matching the empirical multi-frequency covariance matrices of the data. This method is highly robust but offers less physical interpretability [76].

One of emerging computational method is neural networks, especially Convolutional Neural Network (CNN). Several recent studies have explored neural-network architectures for this task, have shown promising performance on simulations [82, 83]. However, there are two most important caveats with usual use of CNN, firstly it does not work in non-Euclidean space and it does not have any reliable uncertainty quantification for scientific use. To address these issues, Recent progress has been made by Perraudin et al. (2019), who developed a spherical CNN framework for cosmological applications known as *DeepSphere* [84]. This method was later extended by Petroff et al. (2020), who introduced a Bayesian spherical CNN, capturing both spectral and morphological aspects of foregrounds [85]. Although their model was trained on temperature maps, as the authors suggested, this approach could in principle be extended further to polarisation maps. In summary, the use of neural network is promising tool, but in cosmological analysis it remains at a developmental stage.

A simplified toy model was constructed to illustrate the basic idea of neural-network-based foreground removal, with further details and limitations discussed in Appendix A.

3.3.3 Prospects for Future B-Mode Detection

Figure 7 summarises the historical evolution of the upper limits on the tensor-to-scalar ratio r, from early *BICEP1/BICEP2* measurements and joint analyses with *Planck*, through to the most recent *Keck Array* constraints. The figure also indicates future targets around $r = 0.001$, a representative benchmark motivated by slow-roll inflationary models and adopted by planned missions such as *LiteBIRD*. A simple exponential fit to these historical limits suggests that $r \approx 6 \times 10^{-4}$ may be attainable by 2040, with a $\sim 76\,\%$ probability under current technological trends.

To complement this forecast, the level of thermal–dust suppression required for a future detection was also assessed. The calculation presented here is carried out independently, but follows the Fisher-information methodology outlined in Dunkley et al. (2009) [90]. The variance of the measured power spectrum is given by

$$\sigma^2(C_\ell) = \frac{2}{(2\ell+1) f_{\rm sky}} \left(C_{\ell,\rm total}\right)^2 \tag{29}$$

with $f_{\rm sky}$ the observed sky fraction and $C_{\ell,\rm total}$ the total residual power. The total signal-to-noise

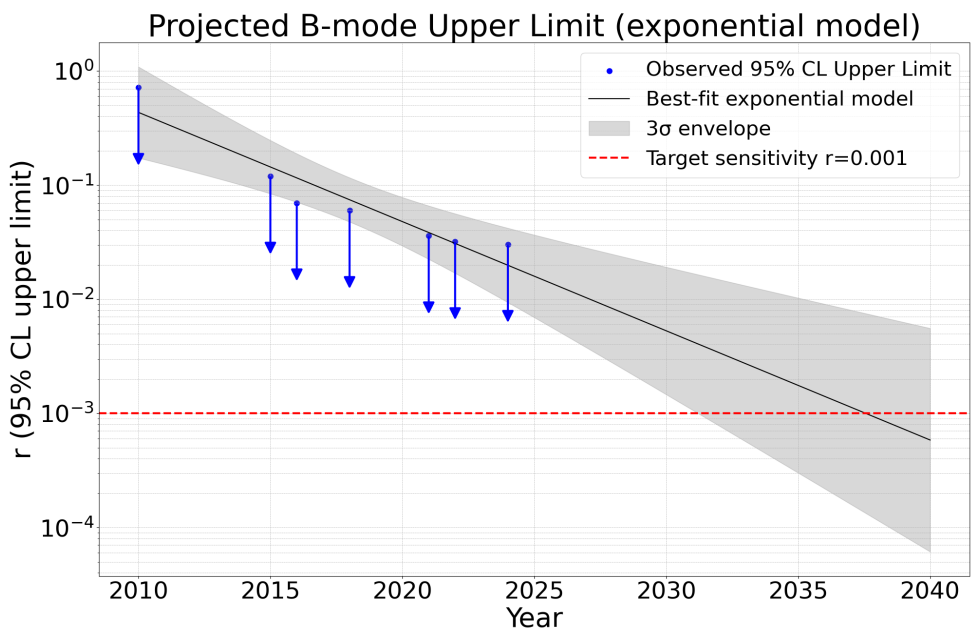

Figure 7: Evolution of the 95% confidence-level upper limits on the tensor-to-scalar ratio r from 2015 to 2024, with a best-fit logarithmic trend extrapolated to 2040. The shaded region represents the $\pm 3\sigma$ confidence envelope in the fitted model. A horizontal dashed line indicates the nominal target sensitivity of $r = 0.001$ illustrating that current extrapolated improvements would not yet reach this threshold by 2040 [61, 64, 86–89].

ratio of the primordial B-mode spectrum is then

$$\mathrm{SNR}_{\mathrm{total}} = \left(\sum_{\ell=\ell_{\min}}^{\ell_{\max}} \left(\frac{C_\ell^{BB\,(\mathrm{prim})}}{\sigma(C_\ell)} \right)^2 \right)^{1/2} \tag{30}$$

where $C_\ell^{BB\,(\mathrm{prim})}$ denotesthe primordial B-mode signal at the target value of r (e.g. $r = 0.01$). As upcoming missions are expected to achieve significant progress in controlling instrumental systematics, delensing, and removal of synchrotron and other foregrounds, only thermal dust was considered as the dominant limiting component. The `PySM3 d1` dust model was used, and the Galactic plane between latitudes $-20°$ and $20°$ was masked, yielding a sky coverage of approximately 65%. Adopting a *LiteBIRD*-like sky fraction of $f_{\mathrm{sky}} \approx 0.7$, it is found that the polarised thermal dust emission must be suppressed by approximately 97 % to achieve a signal-to-noise ratio of $S/N \simeq 3$ for the recombination bump within the multipole range $30 < \ell < 150$.

Although historical trends show steady progress in tightening the upper limits on r, the analysis showed that detecting the recombination bump at meaningful significance requires significant suppression of thermal dust. This emphasises the need for broad-frequency satellite observations and a better understanding of polarised dust emission.

4 Conclusion

Inflation predicts both scalar and tensor perturbations, with the latter producing the primordial B-mode signal. The scalar perturbations, responsible for temperature anisotropies and E-mode polarisation, are now strongly supported by precise CMB observations. In contrast, the tensor perturbations remain undetected due to many limitations such as lensing, instrumental noise and foreground objects which produce B-mode polarisations as well. Although systematic errors in modern instruments are now considered to be well controlled, foregrounds and lensing induced B-modes remain the dominant obstacles.

For delensing, by combining internal reconstruction and cross-correlation, recent work achieved 60 % reduction in the uncertainty of tensor-to-scalar ratio r. Future missions such as *CMB-S4* and *LiteBIRD* forecast promising outlook for future delensing capabilities.

However, foreground emissions remain the major obstacle, as they exhibit highly complex and spatially varying patterns across the sky, closely linked to the Galactic magnetic field. The B-mode amplitude from galactic dust emission reaches approximately $0.01\,\mu\mathrm{K}^2$ over $40 < \ell < 120$, coinciding with the multipole range of the primordial recombination bump. Consequently, polarised galactic dust emission constitutes the most significant astrophysical foreground emission, limiting the detection of the primordial B-mode signal.

Recent machine-learning approaches, particularly CNN-based methods, showed promising ability to remove foreground contamination in simulations. However, they currently lack the geometric suitability and uncertainty quantification required for precision cosmology. Although spherical and Bayesian CNNs offer improvements, they remain at an early stage.

Historical trends suggest that reaching $r \sim 10^{-3}$ by 2040 is achievable, but signal-to-noise ratio analysis showed that thermal dust must be suppressed by $\gtrsim 97\,\%$ for a meaningful detection in a *LiteBIRD*-like experiment. This highlights both the importance of broad-frequency satellite measurements and the need for better understanding of polarised dust emission. Further details and limitations are discussed in Appendix A.

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A Neural Network Discussion

A.1 Motivation and Overview

Independent analyses are valuable in scientific research, as they provide additional statistical confidence in the results and conclusions. Among the various approaches to CMB foreground removal, one emerging method is the use of neural networks. Neural networks are particularly promising because they are inherently capable of handling non-linear relationships, guaranteed by the universal approximation theorem [91]. This contrasts with most traditional component-separation methods, which typically rely on linear assumptions. Such property is particularly advantageous in this context, as foreground emissions exhibit highly complex characteristic, making them difficult to model accurately. Neural network based approaches therefore provide a complementary form of analysis and can serve as a valuable cross-validation tool alongside conventional techniques.

A neural network is a machine learning model composed of interconnected layers of artificial neurons. Each neuron receives weighted inputs, adds a bias, and applies an activation function to produce an output. When arranged in multiple layers, these neuron can approximate complex mappings between inputs and outputs. The network's parameters are optimised through back-propagation using gradient-based algorithm such as stochastic gradient descent that iteratively minimise a chosen loss function.

Among various neural network architectures, the Convolutional Neural Network (CNN) is a particularly suited for analysing data with spatial structure such as images. CNNs employs small filters (kernels) that slide across the input to perform convolutions, producing feature maps that capture local spatial patterns. This makes them highly effective for images processing tasks such as denoising, segmantaion and foreground removal in CMB polarisation maps.

A.2 Toy Model Demonstration

To provide a concrete illustration of how a neural network might be applied to foreground removal, a simplified toy model was constructed. The purpose of this simulation was not to

achieve physical accuracy but to highlight the key challenges that arise when attempting to recover primordial B-mode structure in the presence of dominant foreground contamination.

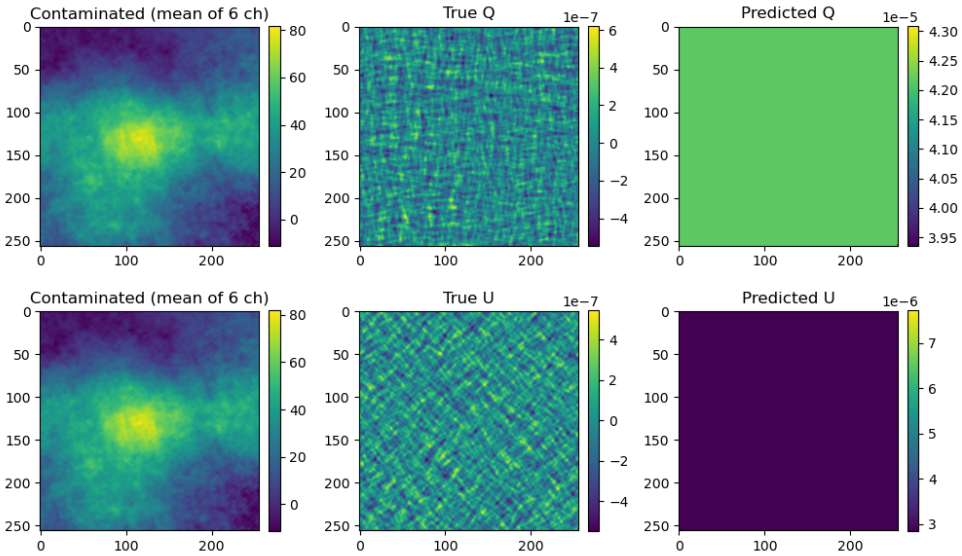

Figure 8: Comparison between the dust-contaminated input maps, true polarisation maps, and neural-network predictions. The left panels show the averaged contaminated maps (mean of six frequency channels), the middle panels show the true Stokes Q and U components, and the right panels display the corresponding network-predicted Q and U maps.

In this simulation, only primordial polarisation and dust contamination were considered. A $10^\circ \times 10^\circ$ patch with 256×256 pixels was used under the flat-sky approximation. Using `Healpy` library provided with python language, primordial E, B-mode spectra were generated with $r = 0.01$ and standard ΛCDM parameters. The 1D spectra $C_\ell^{EE,BB}$ were mapped into 2D $\vec{k}$-space with random amplitudes and phases to represent isotropic quantum fluctuations. They were enforced to be Hermitian symmetry to ensure real-valued maps in real space. These were converted to $\tilde{Q}$ and $\tilde{U}$ via the spin-2 relation and inverse-transformed to real-space Q, U maps.

Dust polarisation was added using `PySM3` with the `d1` model provided in the library. It was centred on the Galactic plane, with 90, 150, and 220 GHz frequency channels. This process was repeated to generate 700 sets of primordial polarisation maps across the three frequencies.
The neural network followed a U-Net architecture, taking six input channels (three frequencies for Q and U) and outputting cleaned Q, U maps in real space. Figure 8 shows examples of the pure primordial maps (the second column) and their counterparts contaminated by dust polarisation (the first column). The network was trained using an L_1 loss function and the Adam optimiser within the `PyTorch` framework.

The third column of Figure 8 is one of the example of output from the model. The model

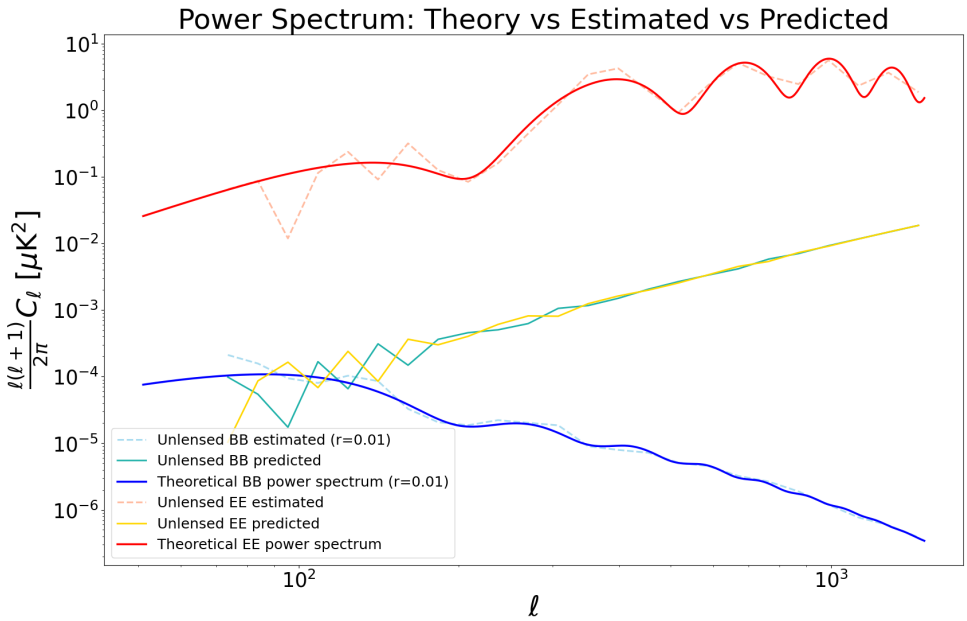

Figure 9: Comparison between the theoretical, re-estimated, and neural network predicted CMB polarisation power spectra. The re-estimated spectra retain the characteristic features of the theoretical curves, indicating that the simulations and flat-sky patch calculations reproduce the expected behaviour. In contrast, the predicted spectra exhibit no such structure, demonstrating that the neural network failed to capture the key statistical properties of the polarisation fields.

reproduced the overall amplitude but failed to recover spatial structures, underfitting due to the vast amplitude disparity between dust and primordial signals. Figure 9 illustrates the theoretical power spectra, re-estimated power spectra and recovered power spectra from output produced by model. For a representative primordial map and its corresponding predicted map, the power spectra were estimated by transforming each map into Fourier space and averaging the squared amplitudes in bins of equal $|\vec{k}|$, assuming $|\vec{k}| \approx \ell$ under the flat sky approximation. The recovered power spectra lacked features such as the acoustic peaks in C_ℓ^{EE} and the reionisation bump in C_ℓ^{BB}. This indicates that the dominant difficulty lies in the extreme dynamic-range difference, suggesting the need for a deeper network.

A.2.1 Current Progress and Challenges

In the toy model demonstration, while the network reproduced the overall amplitude, it failed to recover the spatial structure of the primordial signal, demonstrating that the dominant difficulties arise from the extremely faint nature of the primordial B-modes compared with the much stronger foreground emissions. The typical amplitude of the polarised primordial signal is on the order of $10^{-6}\,\mu$K, whereas the foreground contamination is typically around $10\,\mu$K. The difference of order makes the model numerically unstable and hard to train.

However, in recent works with increasing resolution and model complexity, neural-network ap-

proaches have shown promising performance on simulations, as several studies suggest. For instance, Yan et al. (2023) developed CMBFSCNN (Cosmic Microwave Background Foreground Subtraction with CNN), a CNN inspired by the needlet internal linear combination method [82]. In their work, lensing effects were neglected. The model operates on flat sky patches and was trained on simulated data that included both foreground contamination and instrumental noises to recover primordial polarised signals. The real observational data from *Planck* were divided into twelve equal flat-sky patches for evaluation on their models.

While the reported results appear plausible, there are several caveats to this approach. Firstly, dividing the sphere into flat patches leads to a loss of information near patch boundaries. This problem could be resolved adopting other structure such as transformer based neural network. In their subsequent work, Yan et al. (2025) employed a Transformer-based architecture that handled the curvature of partial-sky maps by representing the sky as a sequence of tokens [83]. However, this still relies entirely on the model's ability to learn spherical features implicitly. Moreover, neither of their models provided statistically meaningful uncertainty estimates, making it difficult to interpret their results in a cosmological context.

These two, operating in non-Euclidean space and the lack of uncertainty quantification, are the most common challenges when applying neural network to physical data. Recent progress has been made by Perraudin et al. (2019), who developed a spherical CNN framework for cosmological applications known as *DeepSphere* [84]. This method was later extended by Petroff et al. (2020), who introduced a Bayesian spherical CNN, capturing both spectral and morphological aspects of foregrounds [85]. Although their model was trained on temperature maps, it produced results comparable to *Planck* temperature observations, while also providing reliable uncertainty estimates. As the authors suggested, this approach could in principle be extended further to polarisation maps, though the much fainter B-mode signal presents an additional challenge.

Apart from architectural limitations, another major challenges lies in preparing appropriate training datasets. To train such a model, one needs labelled training data. However, since the true primordial B-mode signal is unknown, the training must rely on simulated maps constructed from existing foreground models (e.g, `PySM3`), which inherently constrain the model to reproduce previous assumptions rather than discover new physical information. While such networks may improve efficiency of extracting information, they are not fundamentally capable of revealing new information beyond their training priors.

One potential direction to address this limitations involves the use of generative AI to produce more diverse and realistic training data. Although generative models are still dependent on their input distributions, they can combine and randomise features non-linearly, producing greater diversity than classical simulations. Several preliminary studies have explored this approach using Generative adversarial network (GANs), though current implementations remain limited

to small size map rather than full sky [92, 93].

In summary, the use of neural network in cosmological analysis remains at a developmental stage, constrained by the demand for high precision and statistically meaningful uncertainty estimation data. Plus, preparing training datasets is remained challenging. Nevertheless, the inherent capacity of neural networks to model complex relationships continues to make them promising tool for future attempts to detect B-mode.